

A Literature-Implied Negative Answer to a Cuth–Johanis Question

Automatic Math Discovery Run: FA/Banach 001

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Source question

The source paper is Cuth–Johanis, *Isometric embedding of ℓ_1 into Lipschitz-free spaces and ℓ_∞ into their duals*, arXiv:1604.04131. The relevant question appears on page 2:

2

MAREK CÚTH AND MICHAL JOHANIS

Question 2. Let M be a uniformly discrete metric space. Does $\mathcal{F}(M)$ contain a subspace isometric to ℓ_1 ?

In particular, for a few concrete metric spaces for which the answer is not known to us see Remark 10.

Even though we do not know the answer to Question 2, we show that $\mathcal{F}(M)$ is never rotund, which supports the conjecture that the answer to Question 2 is positive. Recall that a normed linear space X is rotund if $\|x + y\| < 2$ for any $x, y \in S_X$, $x \neq y$.

Proposition 2. *Let M be a metric space of cardinality at least 3. Then $\mathcal{F}(M)$ is not rotund.*

This result is interesting also for finite metric spaces: No Lipschitz-free space over an ultrametric space is isometric

Status of this packet

This packet is filed as a literature-implied answer, not as a new original pipeline proof. Ostrovskii later proved a necessary and sufficient criterion for a transportation cost space, equivalently a Lipschitz-free space, to contain an isometric copy of ℓ_1 ; see arXiv:1910.03625. The criterion gives a short negative answer to Question 2 by one of the uniformly discrete examples listed in the Cuth–Johanis paper.

The later criterion

We use the following theorem of Ostrovskii. A sequence of disjoint pairs (x_i, y_i) in a metric space has the *minimum matching property* if, for every n , the edges $\{x_i y_i : 1 \leq i \leq n\}$ form a minimum-weight perfect matching on the vertices $\{x_i, y_i : 1 \leq i \leq n\}$.

Theorem (Ostrovskii). *Let M be an infinite metric space. The Lipschitz-free space $\mathcal{F}(M)$ contains a subspace isometric to ℓ_1 if and only if M contains a sequence of disjoint pairs with the minimum matching property.*

A uniformly discrete counterexample

Let $M = \{v_n : n \in \mathbb{N}\}$ and define, for distinct i, j ,

$$d(v_i, v_j) = 1 + \frac{1}{\max\{i, j\}}.$$

This is a metric: if $i < j < k$, then

$$d(v_i, v_k) \leq d(v_i, v_j) + d(v_j, v_k), \quad d(v_i, v_j) \leq d(v_i, v_k) + d(v_k, v_j),$$

and the remaining triangle inequality is symmetric. Also $d(v_i, v_j) > 1$ for $i \neq j$, so M is uniformly discrete.

Proposition. *The space $\mathcal{F}(M)$ contains no subspace isometric to ℓ_1 .*

Proof. By Ostrovskii's criterion, it is enough to show that no infinite sequence of disjoint pairs in M has the minimum matching property.

Take any infinite sequence of disjoint pairs. Choose two pairs so that the indices of the second pair are larger than the indices of the first pair; after renaming their four vertices, write the indices as

$$q_1 < q_2 < q_3 < q_4.$$

The two original edges have total weight

$$d(v_{q_1}, v_{q_2}) + d(v_{q_3}, v_{q_4}) = 2 + \frac{1}{q_2} + \frac{1}{q_4}.$$

The crossed matching has total weight

$$d(v_{q_1}, v_{q_3}) + d(v_{q_2}, v_{q_4}) = 2 + \frac{1}{q_3} + \frac{1}{q_4}.$$

Since $q_2 < q_3$, the crossed matching is strictly lighter. Thus the original two-edge matching is not a minimum-weight perfect matching on these four vertices. Consequently no infinite sequence of disjoint pairs in M satisfies the minimum matching property, and $\mathcal{F}(M)$ does not contain an isometric copy of ℓ_1 . $\square$

Verification code

The file `code/verify_metric_and_matching.py` checks, on finite initial segments, the triangle inequalities and the crossing inequality used in the proof. These computations are not a substitute for the proof above; they are included only as a reproducibility check for the chosen metric.